

STARD-Net: SpatioTemporal Attention for Robust Detection of Tiny Airborne Objects from Moving Drones

Md. Hasibur Rahman and Sanjay Madria | Department of Computer Science, Missouri University of Science and Technology | Rolla, MO

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Paper & Code

Motivation and Challenges

In shared and contested airspace, moving drones must detect tiny airborne objects in real time to support collision avoidance, swarm awareness, and counter-UAV defense.

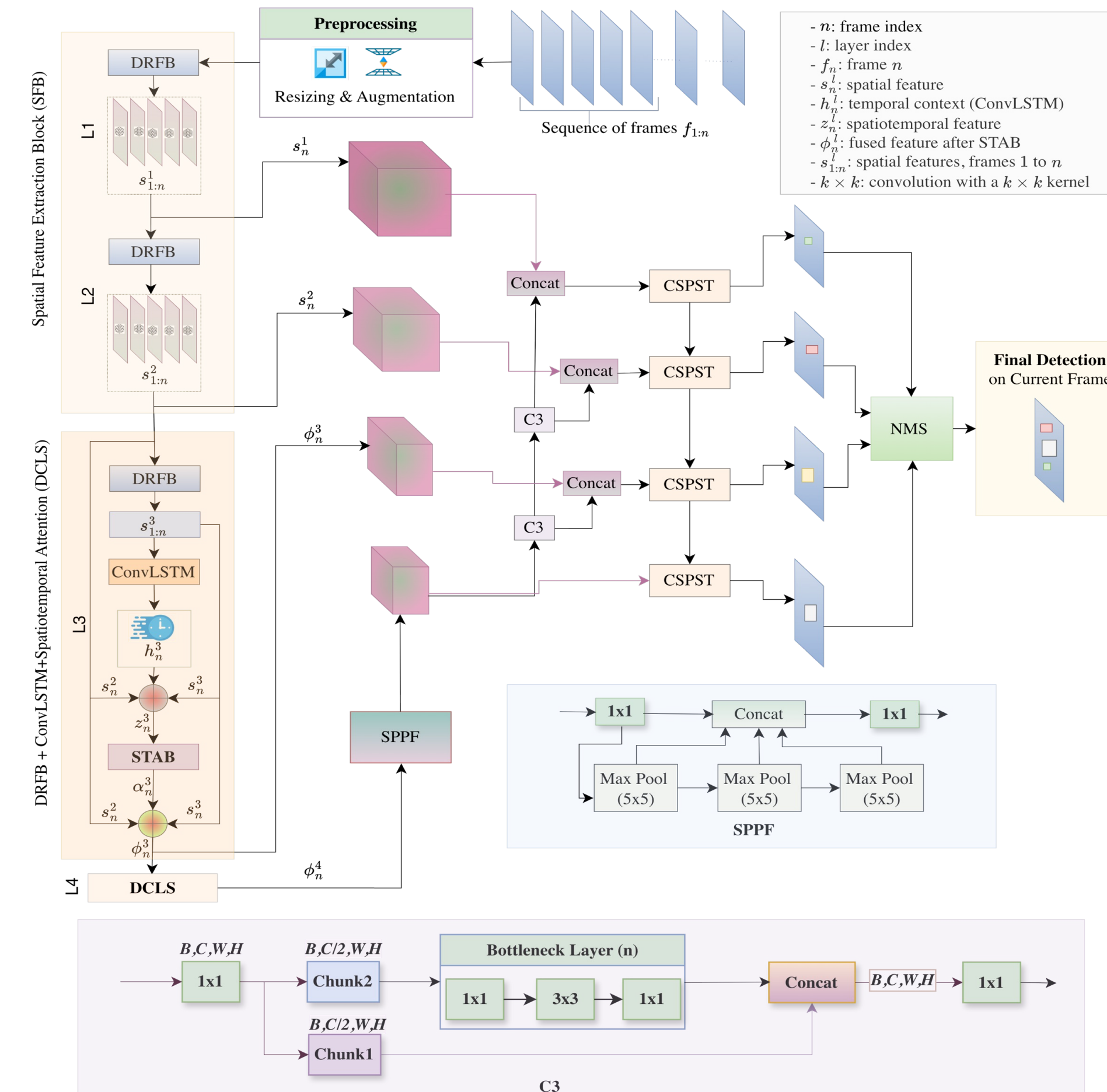


Drone-to-airborne object perception is difficult because both platforms move, making tiny targets blurred, unstable, camouflaged, and weakly visible. **Red** denotes drones, **green** balloons, and **yellow** kites.

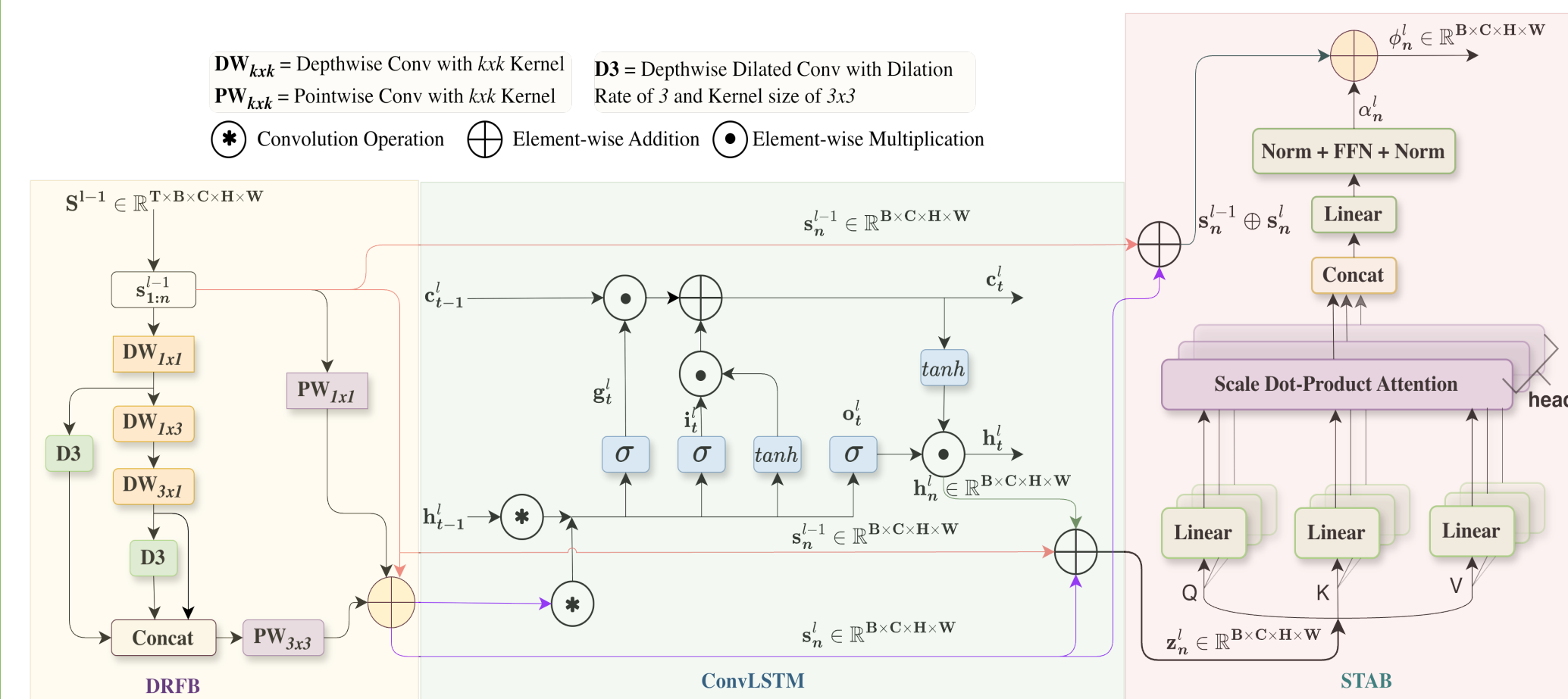
- Targets can occupy only $\sim 0.05\text{--}0.07\%$ of the frame
- **Gap in prior work:** prior detectors often miss tiny airborne targets due to weak single-frame cues, late temporal fusion, or brittle coarse region selection.

Proposed Methodology

Preserve spatial micro-structure early, reinforce it with short-term temporal memory, and suppress temporally inconsistent clutter before the detector.



DCLS block: DRFB preserves spatial detail, ConvLSTM models motion, and STAB refines features with spatiotemporal attention.



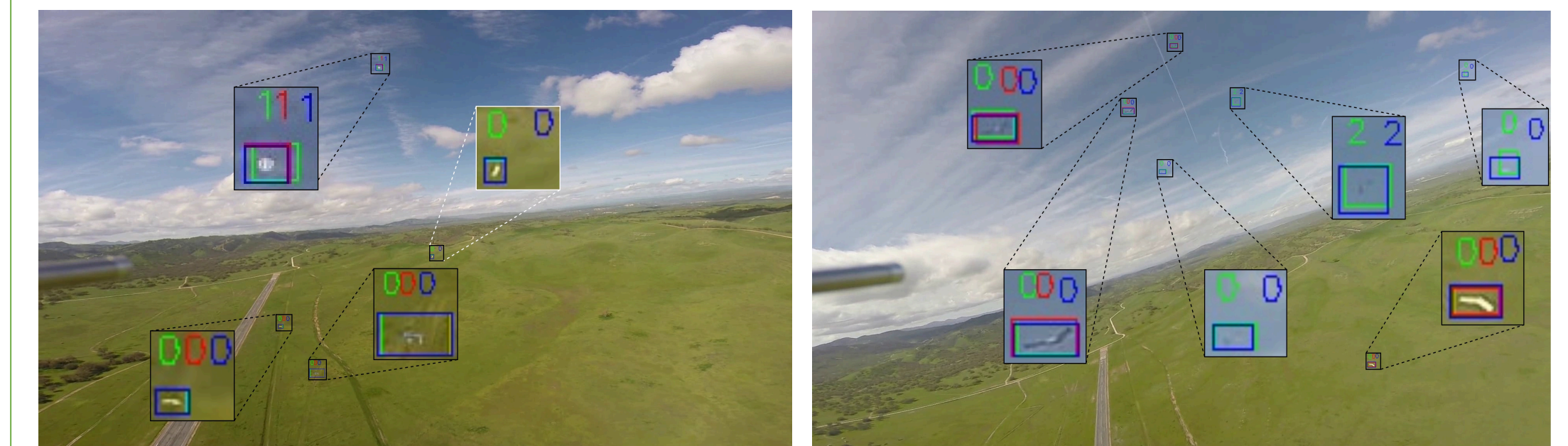
Experiments and Evaluations

Quantitative Comparison Across Datasets

Method	MVAAOD mAP@50	NPS AP@50	FL AP@50	AOT AP@50
TPH-YOLOv5 (ICCVW'21)	0.69	0.90	0.70	0.78
YOLOv12I	0.74	0.93	0.77	0.82
TransVisDrone (ICRA'23)	0.66	0.95	0.75	0.80
C2FDrone (ICRA'24)	0.71	0.93	0.84	0.82
STARD-Net Spatial (Ours)	0.77	0.95	0.78	0.83
STARD-Net Full (Ours)	0.81	0.97	0.81	0.85

Ablation on MVAAOD: ConvLSTM adds temporal context, while STAB suppresses clutter and improves mAP@50 with real-time speed.

Feature Extractor	P	R	F1	mAP @50	FPS
DRFB	0.84	0.72	0.78	0.77	20
DRFB + ConvLSTM	0.82	0.73	0.77	0.76	19
DRFB + ConvLSTM + STAB	0.83	0.77	0.80	0.81	17



Classes: 0 = drone, 1 = balloon, 2 = kite; boxes: **green** = ground truth, **red** = spatial-only detection, **blue** = spatiotemporal detection.

Conclusion: STARD-Net enables real-time tiny airborne-object detection from moving drones, with future work extending it to onboard multi-sensor perception in real-world missions.

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